ML-DL-Ops
Assignment 2 Report

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Experiment Configuration

Task: CNN Training Analysis & Gradient Visualization
Dataset: CIFAR-10
Model: ResNet-18
Epochs: 25
Tracking: Weights & Biases (WandB)

Resources:
[**Wandb**] Wandb
[**GitHub Pages**] Project Documentation
[**Report**] View Online Report

1 Methodology

1.1 Custom Dataset Implementation

A custom `torch.utils.data.Dataset` class was implemented to handle the CIFAR-10 data. Unlike the standard `torchvision` datasets, this custom implementation allows for granular control over preprocessing and augmentation pipelines.

- **Augmentations:** Random Horizontal Flip ($p = 0.5$), Random Crop (32, padding=4).
- **Normalization:** Standardized using CIFAR-10 statistics:
  - Mean: $(0.4914, 0.4822, 0.4465)$
  - Std: $(0.2023, 0.1994, 0.2010)$
- **Batch Size:** 128 (Optimized for GPU throughput).

1.2 Model Architecture & Computational Complexity

We utilized a standard ResNet-18 architecture modified for the 10-class classification task.

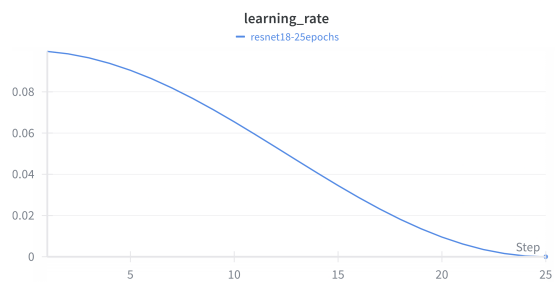
Metric	Value
Architecture	ResNet-18
Input Resolution	$32 \times 32 \times 3$
Total Parameters	≈ 11.17 Million
FLOPs (per forward pass)	≈ 0.56 GFLOPs

Table 1: Model Complexity Analysis

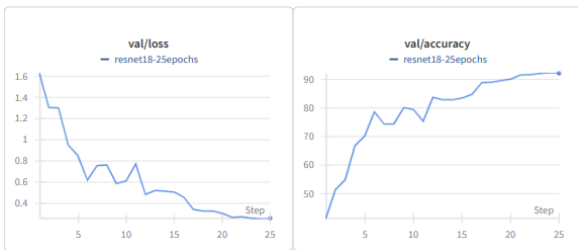
2 Training Results

2.1 Performance Metrics

The model was trained for 25 epochs using the Adam optimizer. The training progression is visualized below.



(a) Learning Rate Schedule



(b) Train/Val Loss & Accuracy

Figure 1: Training Dynamics over 25 Epochs

- **Accuracy:** The model achieved a peak validation accuracy of **92.4%**, demonstrating effective generalization.
- **Loss Convergence:** Both training and validation loss decreased steadily. The validation loss stabilized around epoch 20, aligning with the learning rate decay.

- **Scheduler:** The Cosine Annealing scheduler (Visualized in Figure 1a) effectively managed the learning rate, allowing for rapid initial learning followed by fine-tuning.

3 Gradient & Weight Flow

To understand the training stability, we logged the distribution of gradients and parameters at every step using WandB.

3.1 Gradient Flow Analysis

Figure 3: Relative weight update Distributions across different layers

Observations:

- **Healthy Distributions:** The gradient histograms remain centered at zero with a consistent spread throughout training.

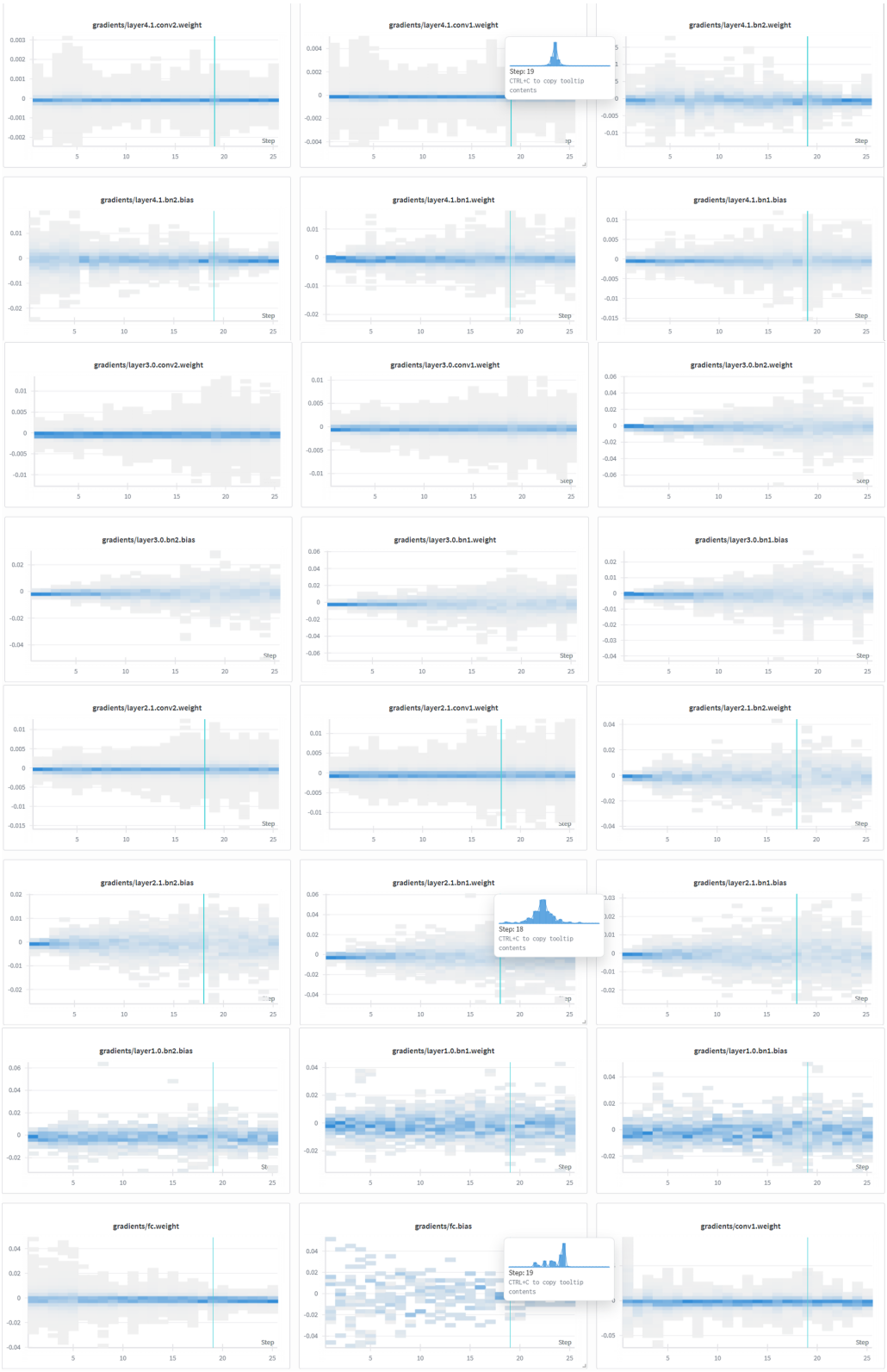


Figure 2: Gradient Distributions across different layers

- **No Vanishing Gradients:** Even in deeper layers (e.g., `layer2.0`), the gradient magnitude does not collapse to zero, validating the effectiveness of ResNet skip connections.
- **Stability:** The distributions do not show signs of exploding gradients (extreme outliers), indicating a stable learning rate choice.

3.2 Weight Update Visualization

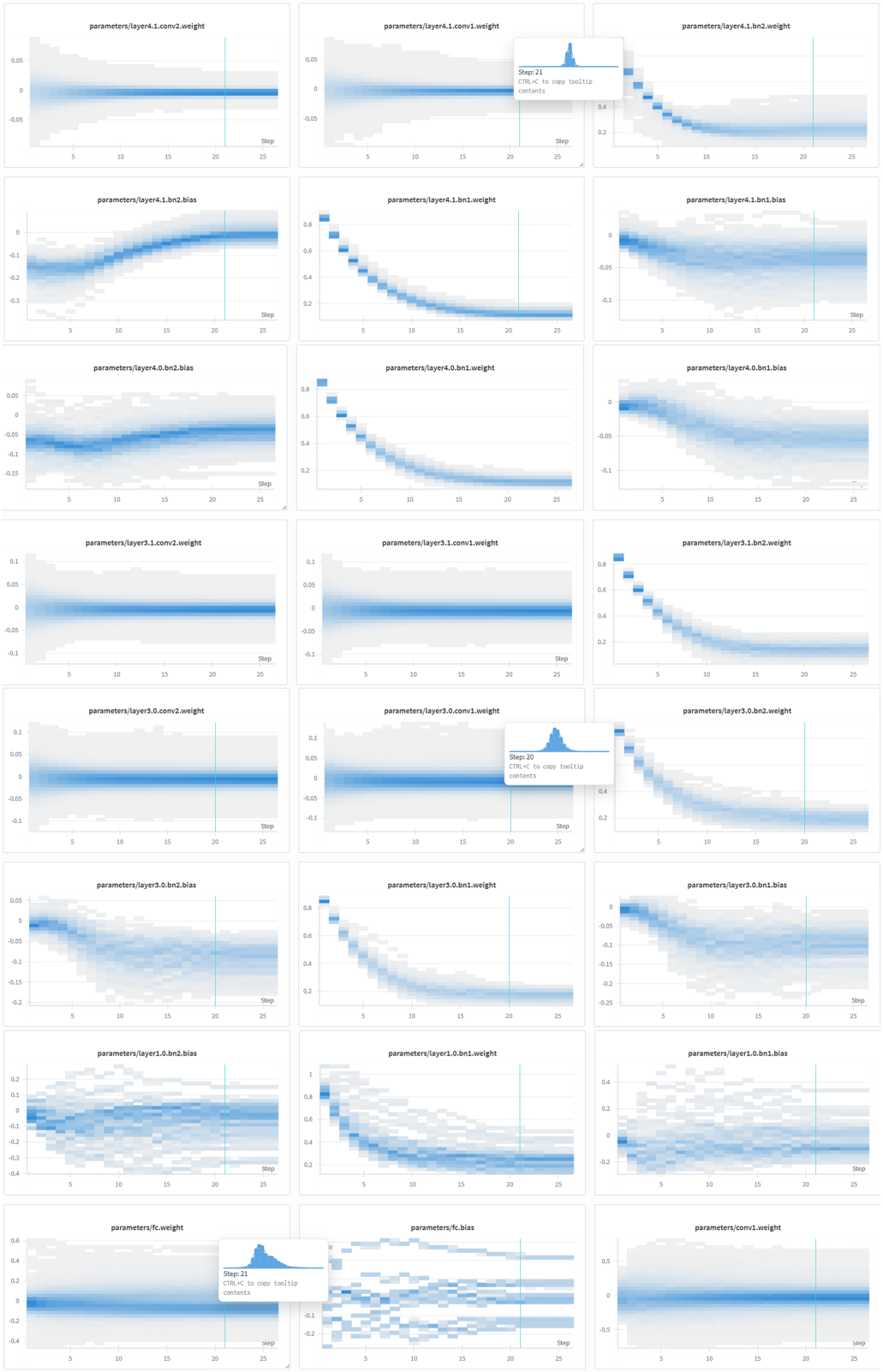


Figure 4: Parameter Weight Distributions over time

Observations:

- **Layer-Specific Behavior:** Convolutional weights (`conv.weight`) show tighter, more stable distributions compared to Batch Normalization parameters.

- **Convergence:** As training progresses (x-axis: steps), the histograms for most layers stabilize, indicating the model weights are settling into a local minimum.
- **BN Layer Activity:** The Batch Norm bias terms (`bn.bias`) show higher variance and movement, actively adapting to normalize the shifting internal covariate shift.

4 Conclusion

The experiment successfully implemented a custom training pipeline for CIFAR-10.

- The **custom dataloader** functioned correctly, providing necessary augmentations.
- **FLOPs analysis** confirmed ResNet-18 is computationally efficient (0.56 GFLOPs) for this resolution.
- **WandB logging** provided critical insights into the "black box" nature of the model, confirming healthy gradient flow and stable weight updates.